BIO 304 . HUMAN ANATOMY & PHYSIOLOGY . MEDMASTERS COLLABORATIVE

Module 3: The Cell

Week 3 teaching guide . 3 topics

This is the instructor-facing companion to the student pre-work hub. For every topic, you have the science to teach, the recommended approach to learning that material, and why it matters to your students' future patients. Lecture order, common misconceptions, and pre-video drawing prompts are baked in so each video lesson can hit the same pedagogical beats.

Topics in this module

1. Cell Structure and Organelles

Plasma membrane, nucleus, and the major cytoplasmic organelles that do the work of the cell.

2. Membrane Transport

Passive vs active, simple vs facilitated, primary vs secondary, plus bulk transport.

3. Cell Cycle and Protein Synthesis

Interphase and mitosis at a glance, then the DNA → RNA → protein pipeline.

Module 3 of 17 . Dr. Sharilyn Rennie

TOPIC 1 OF 3 . WEEK 3 . 7 CARDS (4 DOK 1 / 2 DOK 2 / 1 DOK 3)

Cell Structure and Organelles

Plasma membrane, nucleus, and the major cytoplasmic organelles that do the work of the cell.

The Science

The plasma membrane is the boundary between alive and not-alive at the smallest scale. Phospholipid bilayer with embedded proteins. Selectively permeable: the cell decides what comes in and what goes out. Surface proteins are receptors, channels, transporters, and identity markers. If you teach nothing else about cells, teach the membrane. Everything in physiology happens across membranes.

Organelles divide labor. The nucleus stores DNA. The nucleolus inside makes ribosomal RNA. Free ribosomes make cytoplasmic proteins; rough-ER ribosomes make secreted and membrane proteins. The Golgi modifies and sorts proteins from the ER. Mitochondria are the ATP factories, with their own DNA (legacy of bacterial ancestry) and double membrane. Lysosomes digest debris. Peroxisomes detoxify reactive oxygen species and metabolize fatty acids.

Specialization is the story. A pancreatic acinar cell that secretes massive amounts of digestive enzymes has enormous rough ER. A cardiac myocyte that contracts non-stop is packed with mitochondria. A red blood cell, which just needs to carry hemoglobin, ditches its nucleus and most organelles entirely. Organelle abundance maps onto function; this is a question students will see again and again.

Teaching This Topic

Before the video (drawing prompt)

Ask students to draw a cell with the membrane and label what each major organelle does. They commit, then the video corrects them.

Common misconceptions to address

- Cells are pictured as identical generic blobs. Reality: hundreds of cell types, each shaped for function.
- Mitochondria are 'powerhouses' in cliche. Better: ATP factories, with their own DNA, and they sense and respond to the cell's energy state.
- Students confuse the ER with the Golgi. ER makes; Golgi modifies and ships.

Order of operations (what to teach first)

Membrane first (load-bearing), then nucleus, then the protein-secretion pipeline (ribosome to rough ER to Golgi), then mitochondria, then the cleanup crew (lysosomes, peroxisomes).

Self-test prompts (for students before the recall deck)

- List the organelles a pancreatic acinar cell would have in abundance and why.
- Why are red blood cells unusual in lacking a nucleus?
- What is the difference between rough ER and smooth ER in function?

Why This Matters to Your Patients

Mitochondrial diseases cause exercise intolerance, muscle weakness, and neurodegeneration because high-demand tissues fail first when ATP supply drops. Lysosomal storage diseases (Tay-Sachs, I-cell) accumulate undigested debris in cells, producing progressive organ failure. Drugs that target the plasma membrane, ion channels, receptors, transporters, are the backbone of pharmacology. A patient on a statin has had hepatic mitochondria tweaked. A patient on chemotherapy is having their cell division and protein synthesis machinery hit hard. Cellular biology is the level at which most modern medicine acts.

TOPIC 2 OF 3 . WEEK 3 . 8 CARDS (4 DOK 1 / 2 DOK 2 / 2 DOK 3)

Membrane Transport

Passive vs active, simple vs facilitated, primary vs secondary, plus bulk transport.

The Science

Anything that moves into or out of a cell goes either down its gradient (passive, free) or against its gradient (active, costs ATP). That single distinction organizes the entire topic.

Passive: simple diffusion through the bilayer (small, nonpolar molecules; O2, CO2, steroid hormones). Facilitated diffusion through channels or carriers (polar molecules and ions). Osmosis is water moving down its concentration gradient through aquaporins. All of these are 'free' to the cell. They use the energy already in the gradient.

Active: primary active transport burns ATP directly (the Na/K ATPase is the prototype, pumping 3 Na out and 2 K in per ATP). Secondary active transport uses the gradient set up by primary pumps (the Na-glucose symporter in the small intestine drags glucose in against its gradient by piggybacking on the Na coming in down its gradient). Build the conceptual chain: Na/K ATPase establishes the Na gradient; everything else in the cell uses that gradient as currency. This is why ouabain (a Na/K pump inhibitor) ultimately disables glucose absorption from the gut, even though it never touches the glucose transporter.

Teaching This Topic

Before the video (drawing prompt)

Have students draw a cell membrane and predict which of these molecules cross passively versus require help: O2, glucose, Na, water, cholesterol, K. Then check.

Common misconceptions to address

- Students think active transport means 'fast.' It means 'against the gradient.'
- Students lump all transport into 'osmosis.' Osmosis is specifically water.
- Aquaporins are sometimes thought of as exotic. They are everywhere; without them, water movement would be much too slow for physiology.

Order of operations (what to teach first)

Frame the question (down or up the gradient?), then passive types, then active types, then the Na/K ATPase as the master gradient-builder.

Self-test prompts (for students before the recall deck)

- Without notes, list two molecules that cross by simple diffusion and two that need help.
- Explain why blocking the Na/K pump eventually stops glucose absorption.
- Predict what happens to a red blood cell in pure water.

Why This Matters to Your Patients

Diuretics work by manipulating tubular transporters. Furosemide blocks the NKCC2 cotransporter in the loop of Henle, causing massive Na and water loss. Spironolactone blocks aldosterone (which controls Na reabsorption). IV fluids are designed around osmotic principles: isotonic saline keeps cells stable, hypotonic fluid drives water into cells (treating dehydration), hypertonic fluid pulls water out (treating cerebral edema). Cholera kills by hijacking the intestinal Cl channel, driving massive secretion. Cystic fibrosis is a defective Cl channel. Once you see the body as a network of transport mechanisms, much of clinical pharmacology becomes legible.

TOPIC 3 OF 3 . WEEK 3 . 7 CARDS (4 DOK 1 / 2 DOK 2 / 1 DOK 3)

Cell Cycle and Protein Synthesis

Interphase and mitosis at a glance, then the DNA → RNA → protein pipeline.

The Science

The cell cycle and protein synthesis are two different processes that share importance: both are how a cell builds and renews itself. Teach them in sequence.

Cell cycle: interphase (G1 growth, S DNA replication, G2 final prep), then mitosis (PMAT) and cytokinesis. Checkpoints at G1/S, G2/M, and the metaphase spindle prevent damaged cells from dividing. Loss of checkpoint control is the cellular signature of cancer. Most tumor suppressors (p53, Rb) sit at these checkpoints.

Protein synthesis: transcription in the nucleus reads DNA and produces mRNA. The mRNA is processed (5' cap, poly-A tail, introns spliced out), then exits to the cytoplasm. Translation: a ribosome reads codons three at a time, tRNAs deliver amino acids, peptide bonds form. A single point mutation in DNA can yield a single amino acid change in protein. Sometimes that single change is silent; sometimes it is sickle cell anemia (one Glu-to-Val substitution in beta-globin). The fragility of this pipeline, and its robustness against most insults, is one of biology's deepest balances.

Teaching This Topic

Before the video (drawing prompt)

Ask students to draw the cell cycle as a circle with phases labeled, and separately draw a flowchart from DNA to protein.

Common misconceptions to address

- Students think mitosis is most of the cell cycle. Interphase is roughly 90% of the cycle; mitosis is the brief, dramatic finale.
- DNA, RNA, and protein get mashed together. DNA stores; RNA transports; protein executes.
- Students think cancer is one disease. Cancer is many diseases unified by loss of cell-cycle control.

Order of operations (what to teach first)

Cell cycle (where DNA replication and mitosis fit), then DNA-RNA-protein flow, then point mutations as a way to integrate both ideas.

Self-test prompts (for students before the recall deck)

- In what phase of the cell cycle does DNA replication occur?
- Trace the flow of genetic information from DNA to functional protein.
- Why does a mutation in a tumor suppressor gene predispose to cancer?

Why This Matters to Your Patients

Most chemotherapy targets dividing cells: cells in S or M phase get hit hardest. The predictable side effects (hair loss, bone marrow suppression, mouth sores, GI lining damage) all reflect tissues with rapidly dividing cells. Cancer is checkpoint failure; the modern era of targeted therapy attacks specific failed checkpoints rather than all dividing cells. Genetic counseling is built on understanding how single mutations produce phenotypes. Patient education around chemo side effects is more compassionate and more effective when the nurse can explain why specific tissues fail first.